DEPARTMENT OF MECHANICAL ENGINEERING

Date	21st NOV 2023	Maximum Marks	50
Course Code	ME112AT	Duration	90 Min
Semester	I	CIE-I	

ELEMENTS OF MECHANICAL ENGINEERING

Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	Explain with neat sketches and with Pressure-Volume diagram working of Otto-Cycle IC engine.	10	L2	3
2	With a neat sketch explain: a) Parallel Hybrid Electric Vehicle b) Series-Parallel Hybrid electric vehicle.	10	L2	3
3	With reference to the following observations during a trial on a 4- stroke diesel engine: Cylinder diameter 28 Cm; Stroke of piston 35 Cm; Crank shaft Speed 250 rpm; Brake Load 70 kg; Brake Drum Diameter 3 m; Mean effective pressure 8 bar; Diesel oil consumption 5 litters/hour; Specific gravity of Diesel 0.8; Calorific Value of Diesel 45000 kJ/kg **Determine:** Brake Power, Indicated Power, Frictional Power, Mechanical efficiency, Indicated-Thermal efficiency, Brake - Thermal efficiency.	10	L3	3
4	With neat sketches, explain and derive Velocity ratio and Train Value of simple and compound gear trains with the consideration of four Gears system.	10	L2	3
5a	Compare between constant Pressure and constant Volume cycle IC engines.	5	L2	3
5b	Explain with a sketch the concept of Regenerative braking systems used in HEV	5	L2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	0	0	50	00	00	40	10	00	00	00

Academic year Odd Sem 2023-24

DEPARTMENT OF MECHANICAL ENGINEERING

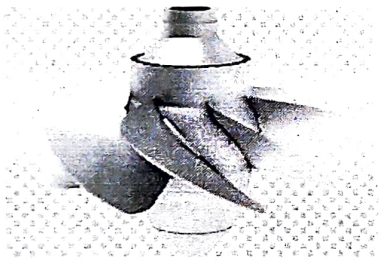
CIE-II

Date	28th December 2023	**Maximum Marks**	50
Course Code	ME112AT	**Duration**	90 Min
Semester	I	**USN**	

ELEMENTS OF MECHANICAL ENGINEERING

Instructions to students:

1. Answer all the Questions.
2. Use of steam tables is permitted.

Sl. No.	Questions	M	BT	CO
1	A rotor of a particular hydraulic turbine is given below in the figure. Guess the corresponding turbine which has such a rotor and with a suitable neat sketch explain the working principle of that turbine. Justify the reason for selecting and explaining this turbine. 	10	L2	3
2	Explain working principle of air standard Brayton cycle with p-V and T-s diagrams clearly stating the thermodynamic processes involved. With a neat schematic diagram apply the same concept for closed cycle gas turbine and justify the working of gas turbine.	10	L3	3
3	With the appropriate block and schematic diagrams, explain the working principle of vapour compression refrigeration (VCR) system. Define COP for the same.	10	L3	3
4	A mixture of saturated water and saturated steam at a temperature of 250 °C is contained in a closed vessel of 0.1 m^3 capacity. If the mass of the saturated water is 2 kg. find the mass of steam in the vessel. Also find the pressure, specific volume, dryness fraction and the enthalpy of the mixture.	10	L3	3
5 a	Briefly explain the formation of steam at constant pressure using temperature enthalpy plot. (Mark all the salient points on the plot).	6	L3	3
5 b	Discuss the differences between impulse and reaction water turbine.	4	L2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	0	0	50	00	00	40	10	00	00	00

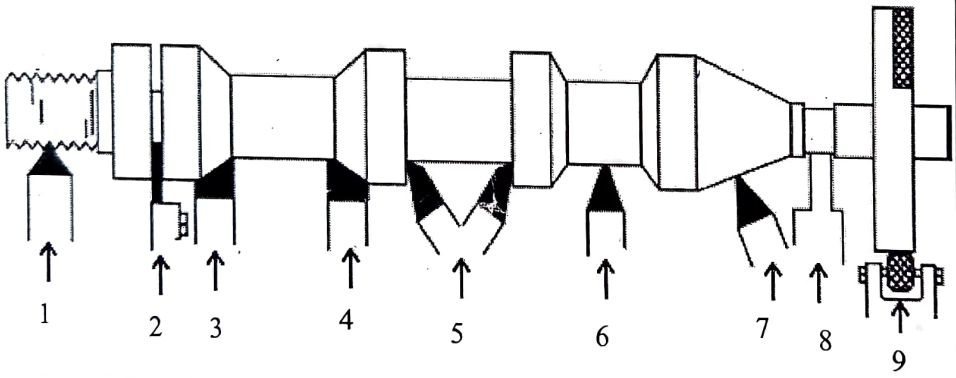
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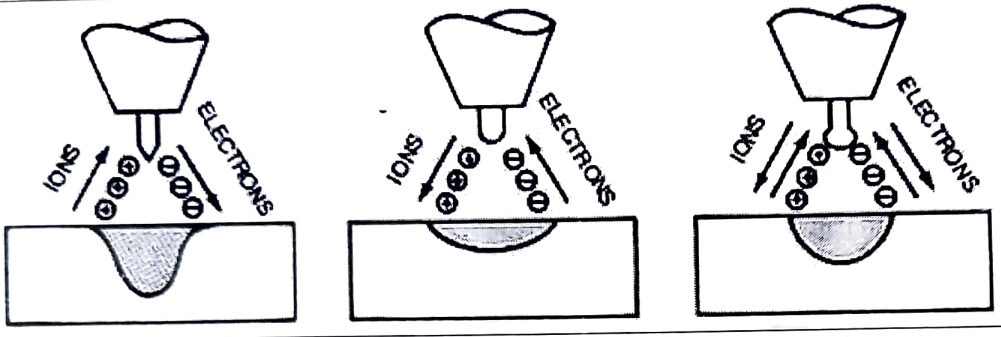
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DEPARTMENT OF MECHANICAL ENGINEERING

Date	23rd January 2024	Maximum Marks	50
Course Code	ME112AT	Duration	90 Min
Semester	I	**IMPROVEMENT CIE**	
ELEMENTS OF MECHANICAL ENGINEERING			

Answer all the Questions.

#	Questions	M	BT	CO
1 a	If you are supposed to buy an engine lathe, then what are the basic specifications that you would consider before buying? With a neat sketch explain the same.	06	2	2
1 b	"CNC machines are better than conventional machines", Justify this statement.	04	3	2
2	The following sketch indicates the different operations performed on a cylindrical workpiece. Identify and explain at least 5 of the operations that are performed clearly. 	10	4	2
3	As an engineer working for a company, you are supposed to test, • the raw material during procurement, • after the machining operation is performed and • finally, during its service. Explain at least one of NDT techniques that you consider in the above different stages and with a neat sketch / block diagram.	10	5	2
4 a	During one of the visits to the welding shop you see the below diagram on display. With your experience, explain the polarities involved and the application for which each of them can be employed.	06	4	1

				
4 b	After getting selected to one of the reputed automobile manufacturing company, your manager informs you that the company wants to develop a new engine having heavy load pulling or off-roading characteristics. You are supposed to select the appropriate engine for the above application. Justify with proper thermodynamic chart which one you would prefer.	04	2	3
5 a	A four cylinder, four stroke spark ignition engine develops a maximum brake torque of 180 Nm at 3200 rpm. Calculate the engine displacement, bore diameter, and stroke length. The brake mean effective pressure at maximum engine torque point is 1150 kPa. Assume bore diameter is equal to stroke length.	06	4	3
5 b	Explain the traction motor characteristics with proper graph.	04	3	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	06	30	14	-	-	10	08	22	10	-

USN | 1 | Q | V | 2 | 3 | M | S | 0 | 4 | 5 |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

I / II Semester B. E. Regular / Supplementary Examinations Feb-2024

ELEMENTS OF MECHANICAL ENGINEERING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 & 9 and 10.
3. Steam table book is permitted.

PART-A

1	1.1	Polyester is an example of _____ type of polymer.	01
	1.2	Define bore of an *IC* engine.	01
	1.3	Connecting rod of *IC* engine is made of _____ material.	01
	1.4	Mention the types of flames in oxyacetylene welding.	02
	1.5	Filler material commonly used in soldering process is known as _____.	01
	1.6	_____ type of turbine is mixed flow turbine.	01
	1.7	Refrigerator works on _____ law of thermodynamics.	01
	1.8	Magnetic particle inspection is carried out for _____ type of material.	01
	1.9	_____ is the equation used to find out the taper angle to obtain taper shape by using swiveling of compound rest method.	02
	1.10	In _____ type of control system, the feedback system is not present.	01
	1.11	The process of reducing diameter of cylindrical workpiece in lathe is called _____ operation.	01
	1.12	Define robot.	01
	1.13	Classify the types of robot based on its configuration.	02
	1.14	In _____ type of steam turbine, the pressure reduces suddenly before entering the turbine.	01
	1.15	Constant volume cycle engines are also called as _____ type of engines.	01
	1.16	Which are the gears that are used to transmit the motion when the shafts are parallel to each other?	02

PART-B

2	a	Differentiate between Thermosetting and thermo plastic polymers.	06
	b	What are composite materials? Classify them & explain the applications of different types of composite materials.	10
3	a	With a neat sketch, briefly explain how to obtain taper turning in lathe machine by using swiveling of compound rest method.	06
	b	How do you carry out the electric arc welding process? Explain with a neat sketch & discuss the general defects in welding products.	10

		OR	
4	a	When do you recommend NDT application during the life cycle of the component? Briefly explain the magnetic particle inspection process with a neat diagram.	10
	b	With a neat diagram explain the working of oxy-acetele welding process.	06
5	a	With a neat sketch explain the working of high head water turbine.	08
	b	Explain with a neat sketch, the Temperature-Enthalpy diagram for formation of steam at constant pressure.	08
		OR	
6	a	With a neat sketch, explain the working of vapour compression refrigerator.	08
	b	With the help of $P-V$ and $T-S$ diagram, explain the working of Brayton cycle for closed cycle gas turbine.	08
7	a	Explain with schematic diagram working principle of IC engine in which burning of fuel takes place at constant volume.	08
	b	The following observations were made during a trial on a $4-S$ diesel engine:	

$Cylinder\ dia\ = 25cm$	$Mean\ effective\ pressure = 6\ bar$
$Stroke\ of\ piston\ = 40cm$	$Diesel\ oil\ consumption = 0.1m^3/min$
$Crank\ shaft\ speed\ = 250rpm$	
$Brake\ load = 70kg$	
$Brake\ drum\ dia = 2m$	
$Specific\ gravity\ of\ diesel\ = 0.78$	
$Calorific\ value\ of\ the\ diesel = 43900kJ/kg$	

Find $BP, IP, FP, \eta_{mech}, \eta_{\beta\ thermal}, \eta_{I\ thermal}$ 08

OR

8	a	With a neat sketch explain series-parallel type of hybrid electric vehicles.	08
	b	With a neat sketch derive the velocity ratio and Train value for simple and compound gear train.	08
9	a	Enumerate mechatronics control system using engine management system as an example with an appropriate diagram.	10
	b	Differentiate between traditional and mechatronics approach of control system.	06
		OR	
10	a	With a neat sketch, explain the anatomy of Robot.	08
	b	Explain with a flow chart the phases of mechatronic system design process.	08